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The Extraction Gap

Redress Delay, Platform Contestability, and Ex-Ante Fiduciary Agency

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The Extraction Gap: Redress Delay, Platform Contestability, and Ex-Ante Fiduciary Agency

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Abstract

Digital platforms can create substantial coordination value while sustaining terms that users find difficult to contest. This paper develops the Extraction Gap, a temporal mechanism linking immediate platform extraction to delayed economic resistance. The model distinguishes individual redress delay from platform sanction delay. Longer individual delay raises the take rate at which contestation becomes privately worthwhile, while longer platform-side delay weakens the present value of deterrence. The paper then introduces an ex-ante acceptance constraint generated by delegated representation. A representative that can search, compare, negotiate, reject, or redirect transactions can improve the user's net executable outside option and tighten the platform's feasible take, provided that refusal is technologically and contractually executable. The effect of AI-based representation is therefore conditional on interoperability, economically meaningful portability, switching feasibility, fiduciary alignment, bounded authorization, and contestability of the agent layer itself. The framework integrates contestation, deterrence, and refusal within a single model of digital-market contestability and derives empirical strategies for separately identifying the effects of redress delay, sanction delay, and ex-ante outside-option improvements.

Keywords: digital platforms; contestability; redress delay; delegated agency; interoperability; consumer protection

JEL classification: D18; D83; L40; L86

1. Introduction

Digital platforms solve genuine coordination problems. They reduce search and transaction costs, match dispersed participants, integrate payments and reputation, and create network benefits that can make markets more efficient and accessible. Yet the same infrastructures can sustain persistent intermediary power. Network effects, switching costs, accumulated data, reputation lock-in, default positioning, and control over complementary services may weaken the competitive pressures that would otherwise discipline platform terms (Rochet and Tirole 2003; Parker, Van Alstyne, and Choudary 2016; Srnicek 2016).

Platforms can create coordination value while controlling the infrastructure through which that value is accessed and distributed. Coordination and extraction can therefore coexist. The analysis draws on both platform economics and

critical accounts of digital intermediation, which illuminate different aspects of the same institutional problem (Zuboff 2019; Srnicek 2016).

Most explanations of persistent platform power focus on structural characteristics of markets or on the forms of value appropriated by intermediaries. These explanations are essential. They leave, however, a further question that is especially important in digital markets: what role does time play in determining whether platform terms are effectively contestable?

A platform may change a commission, ranking rule, access condition, contractual term, advertising load, or data-use practice at the moment a transaction occurs. The economic effect can be immediate. An affected user, by contrast, may need to detect the change, understand its consequences, collect evidence, identify a remedy, submit a complaint, bear legal or administrative costs, and wait for adjudication or recovery. This paper calls the resulting temporal asymmetry the Extraction Gap.

The concept does not denote platform rent itself, nor does it imply that every platform charge is extractive or every delay inefficient. It isolates a narrower mechanism: the interval between the transfer of value and the point at which resistance becomes economically effective.

The analysis distinguishes two different delays. The first is the individual redress delay, τ_i: the time between extraction and economically effective recovery for the affected individual. The second is the platform sanction delay, τ_P: the time between extraction and the point at which successful enforcement imposes an economically relevant consequence on the intermediary. These delays need not be identical, and they operate through different channels.

The individual's decision to contest depends on the discounted expected value of recovery. When recovery is costly, uncertain, and delayed, a larger harm is required before contestation becomes privately worthwhile. The model therefore derives a no-contestation ceiling, θ_c, that rises with individual redress delay. Platform deterrence is distinct: delayed sanctions reduce the present-value marginal deterrent $D_P(\theta)$, so credible contestation need not by itself constrain platform conduct.

The paper then introduces an ex-ante mechanism. A representative that can search, compare, negotiate, reject, or redirect transactions may improve the user's net executable outside option and create an acceptance ceiling θ_A before extraction occurs. Under sufficiently strong deterrence, the sustainable platform take is bounded by $\min\{\theta_c, \theta_A\}$.

These mechanisms correspond to three distinct institutional questions: when contestation becomes privately worthwhile, when a credible challenge actually deters the intermediary, and when unfavorable terms can be refused or redirected before the transaction occurs. The analysis keeps these channels separate throughout.

Artificial intelligence is not treated as intrinsically pro-consumer. An AI agent matters only insofar as it changes identifiable economic parameters such as search cost, response time, switching feasibility, bargaining capacity, or the quality of the user's outside option. A capable agent embedded in a closed or conflicted ecosystem may instead create new dependence.

A representative may itself become a gatekeeper if it controls search, identity, authorization, execution, or accumulated user history. The model captures this through an agent-dependence cost that reduces the effective outside option. Delegation constrains platform terms only if the representative itself remains contestable.

The contribution lies in combining mechanisms that are familiar in isolation. Enforcement delay, switching costs, portability, participation constraints, and delegated search already have established literatures. The model connects them within a single platform setting by separating the threshold for ex-post contestation, the deterrent effect of sanctions, and the ex-ante constraint created by executable outside options.

Section 2 positions the contribution in the relevant literatures. Sections 3-5 develop the Extraction Gap, formal model, and fiduciary-representation mechanism. Section 6 maps the mechanisms to institutional cases. Section 7 presents implications, identification strategies, limitations, and conclusions.

2. Literature and Conceptual Positioning

2.1 Platform coordination, market power, and intermediary take

Digital platforms occupy an analytically ambiguous position in information economics and digital-market policy. They can create substantial coordination value by reducing search and transaction costs, matching participants across market sides, integrating payments and reputation, and enabling network benefits that would be difficult to reproduce through decentralized bilateral exchange. At the same time, the infrastructures through which this coordination occurs can generate durable market power. Network effects, scale economies, accumulated data, reputation, default positioning, and switching costs may make access to the coordination layer itself a source of rent.

Platform intermediation is therefore not treated as intrinsically extractive. The relevant distinction is between the value created by coordination and the terms on which control of the coordinating infrastructure permits the intermediary to appropriate part of that value. This places the argument alongside the economics of multi-sided platforms, information-market competition, and institutional work on platform power.

Switching costs and incompatibility are well-established sources of weak competition. Recent work on digital markets further shows that data portability and interoperability can alter switching frictions and strategic incentives. In the model, these mechanisms matter because they shape the outside option available to an individual or representative (Jeon, Menicucci, and Nasr 2023).

2.2 Ex-post enforcement and ex-ante contestability

A second literature concerns the institutional response to persistent platform power. Traditional competition enforcement is predominantly ex post: conduct occurs, authorities investigate, legal standards are applied, and remedies or sanctions follow. In fast-moving digital markets, this sequence may conclude only after market tipping, user lock-in, or substantial value transfer. This timing problem is central to recent debates over the relationship between ex-post enforcement and ex-ante digital-market regulation (OECD 2024).

Recent digital-market regimes consequently employ obligations relating to interoperability, portability, access, defaults, self-preferencing, and user choice alongside traditional antitrust enforcement. These measures alter the conditions under which transactions and competitive choices occur instead of relying entirely on later correction. The narrower question here is how the timing of effective redress enters the private decision to contest an intermediary's terms and, through that decision, the intermediary's sustainable take (OECD 2024, 2026).

2.3 Timing, sanctions, and deterrence

The importance of timing is well established outside the platform literature. Standard intertemporal deterrence models recognize that delayed consequences may be discounted. Recent empirical work by Dušek and Traxler (2026) provides quasi-experimental evidence that the timing of punishment can affect compliance in the context of speeding offences. Their results also caution against a universal claim that delay has the same behavioral effect in every setting.

The Extraction Gap is not a general theory of delayed punishment. It concerns a specific private-redress environment in which an individual must decide whether a recoverable monetary loss is worth contesting when recovery is uncertain, costly, and delayed. Under those assumptions, delay reduces the present value of recovery and raises the extraction level required to make contestation privately worthwhile.

2.4 Switching costs, portability, and credible outside options

A complementary literature studies switching costs, compatibility, portability, and interoperability. These mechanisms affect the credibility of exit and multihoming rather than the value of ex-post redress. Recent work shows that portability and compatibility can alter switching frictions and strategic incentives (Jeon, Menicucci, and Nasr 2023; He, Li, and Zheng 2023). Here they enter as determinants of the outside option available to an individual or representative.

Portability, interoperability, multihoming, and access to alternative intermediaries are treated as determinants of a credible alternative to incumbent terms, not as policy conclusions assumed in advance. Stronger outside options constrain extraction only when users or their representatives can identify, evaluate, and execute them at the relevant point in the transaction.

2.5 Delegated search and agentic markets

Recent research increasingly treats AI-mediated search as a shift in decision authority from users toward software agents that can interpret goals, compare options, and in some settings execute decisions (Rothschild et al. 2026; Kumaki 2026; Teh, Shimao, and Khern-Am-Nuai 2026). Such systems can reduce search and execution frictions, but they also create questions about transparency, incentive alignment, information access, and market structure. The emerging economics of agentic markets likewise emphasizes that lower technical search costs do not automatically eliminate behavioral and informational frictions (Bichler 2026).

The relevant institution is delegated representation, not AI as such. Artificial intelligence matters only insofar as it changes identifiable economic parameters: search cost, response time, information processing, bargaining capacity, switching feasibility, or the quality of the user's outside option. A highly capable agent embedded in a closed or conflicted ecosystem may reduce some cognitive frictions while creating new dependence.

2.6 Fiduciary agency and incentive alignment

Fiduciary approaches to AI ask how delegated systems can be governed so that their objectives remain aligned with the represented individual (Benthall and Shekman 2023). The relevant problem is familiar from principal-agent theory: advice or delegated authority may be distorted by the agent's own incentives. In the present model, fiduciary independence matters because the representative must be able to reject, redirect, or negotiate in the user's interest. If it is paid to steer toward the incumbent, blocked from alternatives, or unable to transfer relevant data or reputation, the outside-option mechanism weakens.

2.7 The gap addressed by this paper

The adjacent literatures establish the key ingredients separately: platform economics explains network effects and switching frictions; deterrence research studies delayed sanctions; portability and interoperability research studies outside options; agentic-market research studies delegated search; and fiduciary-AI work studies alignment. The paper's question is how these mechanisms interact within a single model of platform contestability.

The Extraction Gap is the temporal asymmetry between immediate extraction and economically effective resistance. The formal analysis derives a no-contestation ceiling from delay, contestation cost, recovery probability, and recovery magnitude, then adds a platform-side deterrence condition and an ex-ante representation constraint. Its contribution is the joint structure rather than any single ingredient.

2.8 Novelty relative to adjacent economic models

A natural objection is that the model simply applies familiar deterrence logic to platforms. Becker-style models and later work already show how expected sanctions affect behavior (Becker 1968; Dušek and Traxler 2026). The distinction here is two-sided timing: the individual's discounted recovery determines whether contestation is initiated, while the platform's discounted sanction determines whether that contestation deters additional extraction. These are separate decision problems with separate delays.

Nor is θ_c a switching-cost parameter. Switching-cost and multihoming models explain the cost of leaving, joining, or using multiple platforms and its effect on pricing and competition (Teh et al. 2023; Teh and Wright 2026). By contrast, θ_c is an ex-post redress threshold that can change with τ_i, c_i, p_i, or λ even when exit opportunities are held fixed. Exit friction and redress friction may interact but are analytically distinct.

The ex-ante ceiling θ_A is mathematically a participation constraint. Its distinctive role here is institutional: delegated, executable choice can alter the outside option before the transaction is completed. Placing that constraint alongside the ex-post contestation threshold and the platform deterrence condition makes it possible to identify which mechanism binds under different conditions.

Accordingly, the novelty claim is integrative rather than primitive: the paper provides a platform-specific Contest-Deter-Refuse framework with distinct timing, binding regimes, and empirical implications.

3. The Extraction Gap

3.1 From market power to temporal asymmetry

Platform power is usually analyzed through structural characteristics: network effects, switching costs, economies of scale and scope, control over data, reputational lock-in, default positioning, and restricted access to complementary services. These mechanisms remain fundamental. The Extraction Gap identifies a different dimension: the asymmetry between the speed at which value can be extracted and the speed at which an affected individual can make resistance economically effective.

Let t_E denote the point at which extraction becomes effective and t_R the point at which individual redress becomes economically effective. Define the individual-side redress delay as:

$$\tau_i = t_R - t_E. \quad (1)$$

The concept does not imply that every platform charge is extractive or that every delay reflects misconduct. It concerns the conditions under which an intermediary can impose terms above a relevant benchmark while the user's ability to contest those terms remains delayed.

3.2 Formal rights and economically effective rights

Suppose an individual possesses a legally valid right to contest a transaction. The existence of that right does not imply that it exerts the same disciplinary effect at every point in time. If successful recovery occurs only after delay, its present value is lower:

$$PV(R) = \beta_i^{\tau_i} p_i R. \quad (2)$$

Effective redress depends on the delay, private contestation cost, success probability, and recoverable amount as well as on the formal existence of a remedy. The same legal right can thus generate different economic incentives across institutional environments.

3.3 Individual redress delay and platform sanction delay

A second temporal process concerns the platform. Even after a user initiates a successful challenge, restitution, penalties, compliance orders, litigation costs, or other consequences may occur only later. Let t_S denote the point at which the platform bears the effective economic consequence of enforcement and define:

$$\tau_P = t_S - t_E. \quad (3)$$

The two delays need not be equal. Individual delay affects whether contestation is privately worthwhile, while platform delay affects the present value of expected sanctions. The Extraction Gap is accordingly represented by the pair (τ_i, τ_P), rather than a single delay parameter.

3.4 Why temporal asymmetry matters in platform markets

The Extraction Gap may be especially consequential in platform markets because extraction is often automated, individual harms may be small relative to the cost of challenge, and platform relationships are repeated. A delay in

effective redress can therefore affect more than one historical transaction: the same contractual or algorithmic condition may continue to govern subsequent transactions while redress remains ineffective.

The relevant analytical question is not simply whether regulation is labelled ex ante or ex post. It is whether the institutional arrangement changes the timing at which users can credibly resist unfavorable terms and the timing at which platforms internalize the consequences of imposing them.

3.5 Distinguishing the Extraction Gap from adjacent mechanisms

The Extraction Gap is not identical to monopoly power. Market power determines the feasible range of terms an intermediary can impose; temporal contestability influences how much of that feasible range can be exercised before challenge becomes privately worthwhile or economically consequential. It is also distinct from switching costs. Switching costs affect the ability to leave, whereas redress delay affects the value of contesting terms while remaining in, or after having participated in, a transaction. These mechanisms interact but should not be collapsed.

Nor is the Extraction Gap a general theory of delayed punishment. The claim is narrower: where an individual evaluates whether a recoverable monetary loss is worth contesting, delayed recovery lowers the present value of recovery and, under specified assumptions, shifts the private contestation threshold. This prepares the formal analysis in the next section.

4. A Dual-Delay Model of Redress and Sustainable Platform Take

4.1 Setup

Consider a stylized transaction mediated by a digital platform P and involving an individual i. A completed transaction generates verifiable surplus $S>0$. The platform chooses a take rate $\theta \in [0,1]$, receiving θS while the individual receives $(1-\theta)S$.

Let $\theta_0 \in [0,1)$ denote a reference take rate from which potentially recoverable excess is measured. The benchmark need not represent a uniquely fair price; it may reflect a competitive reference rate, contractual baseline, regulatory benchmark, or another application-specific comparison point. For $\theta > \theta_0$, suppose successful redress restores a fraction $\lambda \in (0,1]$ of the amount extracted above the benchmark:

$$R(\theta) = \lambda(\theta - \theta_0)S. \quad (4)$$

Let $p_i \in (0,1]$ denote the probability that individual redress succeeds, $c_i > 0$ the private cost of contestation, $\beta_i \in (0,1)$ the individual's per-period discount factor, and $\tau_i \geq 0$ the individual redress delay. The individual contests if and only if:

$$\beta_i^{\tau_i} p_i R(\theta) \geq c_i. \quad (5)$$

Substituting (4) yields the threshold take rate:

$$\theta_c = \theta_0 + c_i / (\beta_i^{\tau_i} p_i \lambda S). \quad (6)$$

For an interior case assume $\theta_c<1$. The quantity θ_c is the no-contestation ceiling: the take rate above which challenge becomes privately worthwhile. It is not the platform's optimal take rate.

4.2 Individual redress delay

Proposition 1 (Individual-delay effect). Holding c_i, p_i, λ, S, θ_0, and β_i constant, the no-contestation ceiling is strictly increasing in individual redress delay:

$$\partial\theta_c / \partial\tau_i > 0. \quad (7)$$

Proof. From (6), $\theta_c = \theta_0 + [c_i/(p_i \lambda S)]\beta_i^{-\tau_i}$. Hence $\partial\theta_c/\partial\tau_i = [c_i/(p_i \lambda S)](-\ln \beta_i)\beta_i^{-\tau_i} > 0$ because $0<\beta_i<1$. Equation (6) also implies $\partial\theta_c/\partial c_i>0$, $\partial\theta_c/\partial p_i<0$, and $\partial\theta_c/\partial\lambda<0$. Thus the result concerns the threshold for private contestation, not the platform optimum. □

4.3 Platform sanction delay and deterrence

Let $\tau_P\geq 0$ denote the delay between extraction and the point at which the platform bears the economic consequence of successful enforcement. Let $p_P\in(0,1]$ denote the platform's perceived probability that a credible challenge ultimately produces an economically relevant consequence for the platform, and let $\beta_P\in(0,1]$ be the platform's discount factor. This probability is conceptually distinct from p_i, the individual's perceived probability of obtaining recovery. The two may be correlated and may coincide under a single enforcement channel, but they need not: private recovery, administrative sanction, restitution, and other platform consequences can follow different procedures and success rates. The core comparative statics do not require independence between p_i and p_P.

$$\Pi_P(\theta) = \theta S. \quad (8)$$

For $\theta>\theta_c$, contestation occurs. Let $x\equiv\theta-\theta_c$ denote the amount by which the platform exceeds the contestation ceiling. Let $H(x)\geq 0$ denote the sanction-equivalent loss conditional on an economically consequential enforcement outcome, with $H(0)=0$ and $H'(x)>0$ for $x>0$. This formulation makes the platform payoff continuous at θ_c while allowing sanctions to rise with the degree of excess extraction. Then:

$$\Pi_P(\theta) = \theta S - \beta_P^{\tau_P}\, p_P\, H(\theta-\theta_c), \text{ for } \theta>\theta_c. \quad (9)$$

Define marginal deterrence intensity:

$$D_P(\theta) = \beta_P^{\tau_P}\, p_P\, H'(\theta-\theta_c). \quad (10)$$

4.4 When does the contestation ceiling bind?

Proposition 2 (Binding-contestation condition). Suppose $H(0)=0$, H is differentiable for $x>0$ with $H'(x)>0$, and $D_P(\theta)\geq S$ for every $\theta\in(\theta_c,1]$. Then, absent a tighter ex-ante acceptance constraint, θ_c is a global maximizer of platform payoff and the sustainable platform take is $\theta^S=\theta_c$.

Proof. For $\theta\leq\theta_c$, $\Pi_P(\theta)=\theta S$, so payoff is strictly increasing up to θ_c. Because $H(0)=0$, the payoff defined in (8)–(9) is continuous at θ_c. For $\theta>\theta_c$, $\partial\Pi_P/\partial\theta=S-D_P(\theta)\leq 0$ by assumption. Hence payoff cannot exceed $\Pi_P(\theta_c)$ anywhere above the threshold, while it is strictly lower anywhere below θ_c. Therefore θ_c is a global maximizer. □

4.5 The platform-delay effect

Lemma 1 (Platform-delay effect). Holding p_P, H, θ, and θ_c fixed, a longer platform sanction delay weakly reduces marginal deterrence and strictly reduces it whenever the platform discounts future consequences:

$$\partial D_P/\partial \tau_P = (\ln \beta_P) D_P(\theta) \leq 0, \text{ with strict inequality if } 0<\beta_P<1. \quad (11)$$

The lemma concerns a different object from Proposition 1. A longer τ_i shifts the contestation threshold θ_c upward; a longer τ_P weakens marginal deterrence conditional on a given excess take. Because θ_c itself depends on τ_i, simultaneous changes in both delays can generate cross-effects. Lemma 1 is therefore a ceteris-paribus statement with θ_c held fixed.

4.6 Ex-ante representation and the acceptance ceiling

Let u_0 denote the individual's net outside option without delegated representation. Introduce a representative acting before transaction completion. Let v_A denote the gross payoff from the best alternative that the representative can discover and execute, $k_A \geq 0$ the incremental cost of switching, agency, or execution, and $g_A \geq 0$ the dependency loss created by agent lock-in, conflicted steering, or vertical integration. The ex-ante mechanism applies only where the incumbent transaction requires user or agent authorization before execution and where the representative can credibly refuse, redirect, or withhold that authorization. Define the net executable outside option:

$$u_A^{eff} = v_A - k_A - g_A. \quad (12)$$

The represented individual accepts the incumbent transaction only if:

$$(1-\theta)S \geq u_A^{eff}. \quad (13)$$

The ex-ante acceptance ceiling is therefore:

$$\theta_A = 1 - u_A^{eff}/S. \quad (14)$$

For the interior case used in Proposition 3, assume $0<u_A^{eff}<S$, so $0<\theta_A<1$. If $u_A^{eff} \leq 0$, the implied ceiling is at least one and is non-binding over $\theta \in [0,1]$. If $u_A^{eff} \geq S$, even $\theta=0$ fails the participation condition and no voluntary incumbent transaction occurs. Unlike θ_c, which arises from the attractiveness of contesting after unfavorable terms are imposed, θ_A arises from the feasibility of rejecting or redirecting those terms before the transaction occurs. If the intermediary can impose the relevant term without contemporaneous authorization, or if participation is effectively unavoidable, θ_A need not be a binding constraint; in that case the model reverts to the ex-post contestation and deterrence channels for that transaction.

4.7 The dual-constraint result

With ex-ante representation, the platform solves:

$$\theta^S = \arg\max_{\theta \in [0, \theta_A]} \Pi_P(\theta). \quad (15)$$

Proposition 3 (Dual-constraint result). Assume $0<u_A^{eff}<S$, ex-ante refusal or redirection is feasible, and the strong-deterrence condition $D_P(\theta) \geq S$ holds for every $\theta \in (\theta_c,1]$. Then the sustainable platform take is:

$$\theta^S = \min\{\theta_c, \theta_A\}. \quad (16)$$

Proof. The platform maximizes Π_P over the feasible set [0,θ_A]. If θ_A≤θ_c, payoff equals θS throughout the feasible set and is maximized at θ_A. If θ_c<θ_A, Proposition 2 implies that payoff is maximized at θ_c and cannot be increased by moving above it. Hence in either case θ^S=min{θ_c,θ_A}. If deterrence is weak, θ^S may exceed θ_c; and if refusal is infeasible, θ_A does not restrict the feasible set. □

4.8 The net outside-option effect

Corollary 1 (Net outside-option effect). For an interior voluntary transaction with fixed S, delegated representation tightens the ex-ante acceptance constraint if and only if the net executable outside option under representation exceeds the unrepresented outside option:

$$u_A^{eff} > u_0 \iff \theta_A^{rep} < \theta_A^0. \quad (17)$$

Proof. Without representation, θ_A^0=1−u_0/S. With representation, θ_A^{rep}=1−u_A^{eff}/S. Since S>0, θ_A^{rep}<θ_A^0 if and only if u_A^{eff}>u_0. Thus this is an immediate comparative-static implication of the participation constraint rather than a separate structural proposition. □

4.9 Robustness, boundary assumptions, and scope

Four boundary assumptions matter. First, p_i and p_P describe different outcomes and need not be independent; p_i is the probability of effective individual recovery, whereas p_P is the probability of an economically relevant platform consequence. Second, H(0)=0 makes the platform payoff continuous at θ_c. Third, the effect of τ_P on deterrence is ceteris paribus: because θ_c depends on τ_i, simultaneous changes can generate cross-effects through H'(θ-θ_c). Fourth, θ_A binds only where refusal or redirection is technologically and contractually executable; essential or effectively mandatory transactions may lack this ex-ante constraint.

5. Ex-Ante Fiduciary Representation

5.1 From redress to representation

Individuals can constrain platform terms through two different routes. Ex-post contestation is governed by θ_c, whereas ex-ante representation is governed by θ_A. The distinction is institutional rather than technological. A human adviser, cooperative intermediary, automated service, or AI agent could perform at least some representative functions; AI mainly changes the scale and speed at which individualized representation can be supplied.

5.2 A richer outside-option function

The gross alternative payoff should not be treated as exogenous in institutional interpretation. Let:

$$v_A = v(I, D, M, N, Q, F, N_A, A_R), \quad (18)$$

where I denotes interoperability, D economically meaningful portability of data and reputation, M switching or multihoming feasibility, N the availability and quality of alternative providers, Q search/comparison/execution capability, F fiduciary alignment, N_A effective competition among agent providers, and A_R authorization and revocability quality.

Agent dependency can be represented as:

$$g_A = g(L_A, V_A, X_A), \quad (19)$$

where L_A denotes lock-in to the agent ecosystem, V_A vertical integration with incumbent platforms or suppliers, and X_A switching or exit costs from the agent itself. Combining (18) and (19) with execution cost gives:

$$u_A^{eff} = v(I,D,M,N,Q,F,N_A,A_R) - k_A - g(L_A,V_A,X_A). \quad (20)$$

This expression is the institutional core of the ex-ante mechanism.

5.3 Interoperability and portability

Interoperability matters because an outside option is valuable only when alternative systems can interact with the user, representative, and relevant transaction infrastructure. Portability matters because switching may otherwise require the user to abandon economically valuable data, ratings, transaction history, credentials, or reputation. In reputation-intensive markets, low technical switching costs can coexist with high economic switching costs if accumulated trust cannot move. This interpretation is consistent with formal work showing that data portability and interoperability can alter switching costs and compatibility incentives (Jeon, Menicucci, and Nasr 2023).

The variable D therefore refers to economically meaningful portability rather than mere file export. Likewise, I refers to actionable interoperability: the ability of a representative to access, compare, authorize, or execute across systems under appropriate security and governance conditions.

5.4 Switching, multihoming, and executable alternatives

Switching requires substitution from one provider to another; multihoming permits simultaneous participation. In markets with strong network effects, multihoming may be more important than complete exit because it permits the user to access alternatives without immediately sacrificing the incumbent network. A fiduciary representative can therefore discipline platform terms even when full exit is unrealistic, provided it can redirect marginal transactions toward competing services.

5.5 Capability and fiduciary alignment

AI may substantially increase Q by reducing search cost, continuously comparing offers, processing complex terms, monitoring changes, and executing switching or negotiation. But capability and alignment are analytically distinct. A technically powerful agent that is rewarded for steering users toward higher-margin products, owned by the incumbent platform, or unable to access rival services may fail to improve v_A and may increase g_A. Recent work on delegated digital search and agentic markets similarly treats search, comparison, and execution capability as economically consequential while cautioning that platform architecture and residual frictions can preserve gatekeeping power (Rothschild et al. 2026; Bichler 2026).

Fiduciary status has economic content here. It constrains the representative's objective function and conflicts of interest. The relevant distinction is between dependent intermediation and independent representation, not between human and AI. Fiduciary-AI research provides a governance analogue by treating loyalty and care as constraints on the operator and design of automated systems (Benthall and Shekman 2023).

5.6 Authorization and revocability

Representation also requires authority. An agent that can only recommend does not create the same outside option as one authorized to act, yet broader authority increases governance risk. Effective delegation therefore requires authority that is scoped, auditable, revocable, attributable, and understandable to the represented individual. The objective is bounded, effective authority rather than maximal automation.

5.7 The new-gatekeeper problem

Delegated representation creates a paradox. The institution designed to reduce dependence on incumbent platforms may itself become a new intermediary with substantial market power. If a small number of personal-agent providers control discovery, comparison, authorization, payment routing, and transaction execution, they may become new gateways to market access. The user may become less dependent on the original platform but more dependent on the agent layer. This concern is increasingly explicit in competition-policy analysis of agentic AI (Autorité de la concurrence 2026; Boa, Carpenter, and Dorrell 2026).

The model captures this risk through g_A. Agent-level lock-in, conflicted steering, or vertical integration weaken the beneficial outside-option effect of representation. Delegation constrains platform extraction only when the representative itself remains contestable.

5.8 Competition among agents and open standards

Competition among agents can reduce g_A, but provider count alone is insufficient. If competing agents depend on the same gatekeeper, exclusive data source, model provider, app ecosystem, identity system, or payment rail, nominal competition may coexist with structural dependence. Effective agent competition therefore depends on contestability across several layers, including the interface, models, data access, identity and authorization, and transaction execution.

Open technical standards can raise interoperability and reduce dependency, whereas proprietary standards controlled by a single operator may create entry barriers and reinforce g_A. Standard-setting governance is therefore part of the competition architecture of agentic markets, and open, interoperable standards have become an explicit policy recommendation in this area (Autorité de la concurrence 2026).

5.9 Institutional complementarity

These institutional features are complementary. Interoperability is of limited value if reputation cannot move; portability does little where viable alternatives are absent; fiduciary independence cannot create an executable outside option without authorization; and authority without revocability creates a new source of dependency. The framework therefore treats interoperability, portability, switchability, alignment, authorization, and agent contestability as jointly relevant rather than individually sufficient.

6. Institutional Evidence and Boundary Conditions

6.1 What the evidence can and cannot show

The cases in this section are not causal tests of the Extraction Gap model and were not designed to estimate τ_i, τ_P, θ_c, or θ_A. They serve a narrower purpose: to show that several institutional conditions in the model—open coordination, low-cost access, interoperability, portability, switching feasibility, delegated execution, and agent contestability—already exist in partial form and at scale. They are evidence of mechanism plausibility and boundary conditions, not validation of the full model.

6.2 UPI: coordination without a single proprietary transaction gate

India's Unified Payments Interface illustrates large-scale coordination through a common payment interface spanning hundreds of participating banks. NPCI reported 758 banks live on UPI and about 25.47 billion transactions in October 2026 (NPCI 2026). Its relevance is limited but useful: coordination at scale need not require a single proprietary consumer-facing transaction gate. In the model, such infrastructure may improve interoperability, provider availability, and switching feasibility rather than constituting fiduciary representation itself.

6.3 Pix: low-cost coordination and the benchmark problem

Brazil's Pix provides a second example. More than 150 million individuals used Pix and more than 7.8 billion transactions were completed in September 2026 (Banco Central do Brasil 2026). BIS analysis reports an average merchant cost of about 0.22% of transaction value for Pix, compared with about 2.2% for credit cards and 1.1% for debit cards in Brazil (BIS 2024). These figures do not establish a general benchmark for platform rents because payment systems and platforms perform different functions; they instead illustrate that coordination infrastructure and exclusive control over access to coordination are analytically separable.

6.4 X-Road: distributed interoperability

Estonia’s X-Road illustrates a different institutional mechanism. X-Road provides a standardized distributed data-exchange layer rather than a centralized broker storing all exchanged information. The case does not establish individual data ownership or fiduciary agency. Its narrower relevance is that interoperability can be institutional infrastructure rather than discretionary bilateral permission granted by an incumbent. This matters for agentic markets, where representatives may need to act across multiple systems while preserving bounded access and auditability (X-Road Global 2026).

6.5 The DMA: contestability through portability and interoperability

The Digital Markets Act offers a contemporary example of policy that changes market conditions before disputes are resolved ex post. Its portability and interoperability obligations map to D and I in the outside-option channel, while related switching measures bear on M (European Commission 2026). The DMA does not test the model, but it shows that regulators increasingly treat these variables as determinants of contestability rather than as consumer conveniences

alone. The short post-treatment period, bundled obligations, strategic compliance, and concurrent technology changes also complicate causal evaluation (Botta 2026).

6.6 Agentic AI and executable switching

The UK Competition and Markets Authority's 2026 analysis of agentic AI describes systems that may move from recommendation toward planning and execution across services. That distinction matters for the model: recommendation mainly changes information, whereas executable switching can alter the outside option. The same analysis identifies smart-data schemes, secure digital identity, and interoperability standards as enabling conditions for switching and consumer choice (Competition and Markets Authority 2026).

6.7 The new-gatekeeper boundary condition

Emerging policy evidence also bears on the model's main boundary condition. In July 2026, the French Autorité de la concurrence warned that the platformisation of AI agents could generate risks involving discrimination, visibility, vertical integration, interoperability barriers, and concentration, and recommended portability, interoperability, and open standards across the agentic ecosystem (Autorité de la concurrence 2026). These concerns map directly onto g_A: a technology that improves switching at one layer can create a new bottleneck at another.

6.8 Mapping institutions to model parameters

Institution	Main mechanism	Model relevance	What it does not show
UPI	Shared/open payment coordination	I, N, M	Fiduciary representation
Pix	Low-friction public-interest payment rail	θ_0 context; I, N	General platform-rent benchmark
X-Road	Distributed standardized data exchange	I	Personal data ownership or AI agency
DMA	Portability, interoperability, switching	D, I, M	Automatic welfare improvement
Personal AI agents	Delegated search and execution	Q, v_A	User alignment by default
Agent portability/interoperability rules	Agent-layer contestability	N_A, g_A	Elimination of gatekeeping

6.9 Boundary conditions

The institutional cases also mark the limits of the framework. Alternatives must actually exist; interoperability cannot create competition where viable providers are absent. Portability must cover economically relevant assets such as reputation, transaction history, credentials, and identity rather than raw file export alone. Delegated authority must be bounded and revocable, agent incentives sufficiently transparent and aligned, and contestability preserved at the agent layer itself.

7. Implications, Limitations, and Conclusion

7.1 From formal rights to effective contestability

A formal right, remedy, or sanction does not have a fixed economic force. Effective contestability depends on when a right becomes usable, when the platform internalizes the consequences of violating it, and whether the individual has a credible alternative before extraction occurs. The framework separates these into contestation (θ_c), deterrence (D_P), and refusal (θ_A).

7.2 Regulatory design

Regulatory effectiveness should not be evaluated only by the severity of available sanctions. Two additional questions matter: how quickly can an affected party make redress economically credible, and how quickly does the intermediary internalize the consequences of successful enforcement? Measures that simplify complaint procedures, reduce filing costs, automate evidence collection, or accelerate dispute resolution primarily affect individual contestability. Measures that accelerate investigations, provisional remedies, compliance orders, restitution, or sanctions primarily affect platform deterrence. The two sets of reforms are complements rather than substitutes.

7.3 Why ex-ante mechanisms are distinct

The ex-ante ceiling θ_A operates before extraction occurs. It arises from the ability to reject, switch, negotiate, or redirect a transaction. This mechanism does not require the user to demonstrate that prior conduct was unlawful or recover a previous loss. Under sufficiently strong deterrence, $\theta^S=\min\{\theta_c,\theta_A\}$. Where $\theta_A<\theta_c$, the ex-ante constraint binds before formal redress becomes relevant. This provides a distinct economic rationale for portability, interoperability, multihoming, delegated search, and executable switching.

7.4 Fiduciary AI is neither necessary nor sufficient

The analysis deliberately avoids the conclusion that artificial intelligence is required for ex-ante representation. A human adviser, cooperative intermediary, automated comparison service, or other representative could perform some of the same economic function. Nor is AI sufficient. The key object is effective delegated agency. AI becomes economically relevant when it changes parameters such as τ_i, c_i, p_i, or the net executable outside option.

7.5 The recursive gatekeeper problem

A representative capable of constraining an incumbent platform may itself acquire substantial control over market access. If preferences, transaction histories, identity credentials, authorization structures, or personalized models accumulate within one agent ecosystem, switching agents may become costly. Any intermediary that improves contestability therefore has to remain contestable itself.

7.6 Welfare implications and limitations

A lower platform take is not automatically equivalent to higher social welfare. Platforms may incur fixed costs, provide valuable infrastructure, improve matching, reduce fraud, invest in innovation, or subsidize one side of a multi-sided

market. Similarly, fiduciary representation involves costs. The model identifies constraints on extraction and bargaining rather than a complete welfare optimum. A full welfare analysis would require endogenous platform investment, innovation incentives, participation decisions, cross-side network effects, and the cost and quality of representation.

Several limitations are material. The framework is partial-equilibrium; θ_0 is application-specific; $H(\cdot)$ is reduced-form; users may have behavioral biases or incomplete preferences; fiduciary alignment is not derived from a full principal-agent model; and the institutional cases establish plausibility rather than causal validation. These limits also indicate where extensions of the model would be most valuable.

7.7 Empirical identification and testable hypotheses

The three mechanisms should be identified separately before they are combined empirically. A cross-sectional correlation between slow redress and high platform take would not identify the Extraction Gap because market power may generate both. The empirical task is to isolate plausibly exogenous variation in redress timing, platform consequences, or the quality of executable outside options.

7.7.1 Testable hypotheses

Hypothesis 1 (individual-side delay). Conditional on platform terms and the expected probability and magnitude of recovery, a longer effective redress delay reduces the probability that an individual contests a transaction. Equivalently, the estimated contestation threshold should rise with delay. This is the most direct empirical implication of Proposition 1 because it can be tested without first establishing that the platform endogenously changes its take rate.

Hypothesis 2 (platform-side delay). Conditional on the expected magnitude and probability of enforcement, longer delay before the platform bears an economically meaningful consequence should weaken behavioral response to enforcement. This hypothesis concerns D_P rather than θ_c and should be tested with firm conduct following enforcement shocks, not with user complaint behavior.

Hypothesis 3 (ex-ante outside option). Reforms that make alternatives more executable—through portability, interoperability, multihoming, or delegated switching—should increase switching or multihoming and, where the ex-ante acceptance constraint binds, reduce the maximum platform take compatible with continued participation.

Hypothesis 4 (agent contestability). The competitive effect of delegated representation should be weaker when users face high costs of switching agents, when agents are vertically integrated with incumbent platforms, or when user history and authorization cannot move across agent providers. This hypothesis operationalizes the agent-dependence term g_A.

7.7.2 Identification strategy for individual redress delay

The preferred design for τ_i exploits a reform or administrative change that shortens effective redress without directly changing platform terms—for example, staggered fast-track procedures, statutory deadlines, or digital dispute-resolution rollout. A difference-in-differences or event-study design can compare affected and unaffected users, products, or jurisdictions. The first-stage outcome should be complaint or contestation behavior: if shorter τ_i lowers the harm threshold for filing, the mechanism is observed before asking whether platforms subsequently adjust terms.

This two-stage approach helps separate the mechanism from reverse causality. A direct estimate of platform take on redress delay is difficult because powerful platforms may both charge more and influence the institutional environment. Showing first that exogenous delay changes contestation, and then testing platform responses, is closer to the model's causal sequence.

7.7.3 Identification strategy for platform sanction delay

Identification of τ_P requires variation in when platforms internalize enforcement consequences, such as reforms accelerating interim measures, restitution, compliance orders, or monetary sanctions. Outcomes should include subsequent platform terms or conduct. Administrative workload or random case assignment may provide instruments for realized delay, but exclusion restrictions must be defended because caseload can correlate with enforcement intensity or case quality.

Where staggered reforms are used, modern heterogeneous-treatment estimators should replace naive two-way fixed effects (Callaway and Sant'Anna 2021; Sun and Abraham 2021; Goodman-Bacon 2021).

7.7.4 Identification strategy for ex-ante contestability

Portability and interoperability reforms provide a distinct design for the ex-ante channel. Useful outcomes include switching, multihoming, alternative-provider use, and changes in incumbent fees or contractual restrictions. The key treatment is the date on which portability or interoperability becomes practically usable, not merely the legal adoption date.

The DMA illustrates both the opportunity and the identification problem. Its obligations may create useful variation, but multiple rules, short post-treatment windows, strategic compliance, and concurrent technology shocks complicate causal attribution (Botta 2026). A credible design should focus on a narrow obligation with an identifiable implementation date and a defensible comparison group rather than on a simple before-after comparison of the DMA as a whole.

7.7.5 A direct field-experimental test of delegated representation

The ex-ante mechanism can also be tested directly through a field experiment that randomizes users among ordinary choice, recommendation-only assistance, and an execution-capable fiduciary agent. Outcomes would include accepted prices or take rates, switching frequency, search time, transaction completion, realized surplus, and rejection of incumbent offers. Additional treatments could vary agent portability, interoperability, or vertical integration to test whether agent dependence g_A offsets gains from improved alternatives. This design provides the closest empirical analogue to θ_A.

7.7.6 Main threats to causal identification

Five threats are especially important: reverse causality, bundled reforms, anticipation, spillovers, and selection into complaints. Preferred designs use narrow implementation discontinuities, within-platform comparisons, pre-specified event windows, direct measures of the relevant delay, and unaffected products or jurisdictions as auxiliary controls where credible. Staggered-adoption designs should use estimators robust to heterogeneous treatment effects. When no

credible source of exogenous timing variation exists, the evidence should be described as mechanism-consistent rather than causal.

7.7.7 Measurement and data requirements

Measurement must correspond closely to the model. τ_i should capture time to economically usable recovery, not administrative acknowledgment; τ_P should capture time to an economically meaningful platform consequence; θ should be operationalized using the relevant fee, commission, contractual burden, or other appropriable term; and the ex-ante channel requires measures of switching feasibility and the net executable outside option.

A credible empirical project would combine platform terms, complaint or dispute records, enforcement timelines, implementation dates for portability or interoperability, switching or multihoming measures, and market-structure controls. These designs are presented as testable implications, not as causal validation drawn from the institutional cases in Section 6.

7.8 Broader implications for digital-market economics

Digital platforms perform two economically distinct functions: they create coordination value and control infrastructures through which coordination occurs. The institutional question is how to preserve the first while keeping the terms of access to the second contestable. One implication of the framework is that contestability depends on the timing and location of representation as well as on competition among firms.

7.9 Conclusion

Digital-platform markets operate at different speeds. Extraction can occur when a transaction is priced, ranked, matched, authorized, or executed; individual redress may become economically meaningful only later, and platform sanctions later still. The term Extraction Gap refers to this temporal structure.

The model identifies two separate delays. The individual redress delay, τ_i, raises the take rate at which contestation becomes privately worthwhile. The platform sanction delay, τ_P, weakens the present-value deterrent associated with future enforcement when platforms discount future consequences. These mechanisms should not be collapsed into a single delay variable.

The ex-ante mechanism adds a separate constraint. A representative that can search, negotiate, reject, or redirect transactions generates an acceptance ceiling, θ_A, through the user's net executable outside option. Under strong ex-post deterrence, $\theta^S=\min\{\theta_c,\theta_A\}$. Together, the three channels are contestation, deterrence, and refusal.

AI can strengthen the third mechanism, but only conditionally. The agent must have access to alternatives, usable portability and interoperability, sufficient delegated authority, aligned incentives, and meaningful independence from the intermediary whose terms it evaluates. And the agent itself must remain contestable.

The claim is narrower than a theory of a new economic system. It is that the timing of redress and the institutional location of representation are economically relevant determinants of platform contestability. When extraction is immediate but resistance is delayed, formal rights may constrain less than their legal existence suggests; institutional

design must therefore address both when rights become usable and whether alternatives can be exercised before value is transferred.

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Author responsibility and AI-assisted preparation note

During preparation of this working paper, the author used OpenAI ChatGPT to support literature-review workflows, manuscript organization, language refinement, consistency checking, and document preparation. All arguments, equations, source verification, revisions, and final judgments were reviewed and approved by the author, who takes full responsibility for the content.

Data availability

No new empirical dataset was generated or analyzed for this theoretical article. The institutional statistics discussed in Section 6 are drawn from the cited public sources.

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